

ABSTRACT

EFFICIENT ALGORITHMS FOR NEAREST CORRELATION MATRIX COMPUTATION WITH MISSING DATA

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This thesis investigates efficient algorithms for computing the **Nearest Correlation Matrix (NCM)** under incomplete financial data. Correlation matrices are vital in portfolio optimization and risk management, yet empirical estimates often violate symmetry, positive semidefiniteness, and unit diagonal conditions due to missing observations. Two projection-based methods are analyzed: the **Modified Alternating Projections (MAP)** and **Anderson Acceleration (AA)**. Theoretical analysis using convex optimization and normal cone characterization supports numerical evaluation on synthetic and real-world stock-return matrices (550×550 , 2020–2025). Missing data are modeled through **Missing Completely at Random (MCAR)** and **Not Missing at Random (NMAR)** mechanisms. The results show that AA converges faster and remains robust at higher missingness levels, preserving the validity of the matrix. The findings highlight Anderson Acceleration as an efficient and reliable tool for estimating valid correlation matrices in financial modeling when datasets are incomplete.

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**EFFICIENT ALGORITHMS FOR NEAREST CORRELATION MATRIX
COMPUTATION WITH MISSING DATA**

BY

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DEDICATION

Dedicated to all the OYHEYINKA — you're not the best, but always remember that you are significant.

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CHAPTER 1

INTRODUCTION

1.1 Background and Context

In quantitative finance, correlation matrices are essential for capturing how assets move together and underpin applications, such as risk management, portfolio optimization, derivative pricing, fraud detection, and regulatory compliance. Harry Markowitz’s seminal paper on modern portfolio theory demonstrated that selecting assets with low or negative correlations reduces portfolio risk without sacrificing expected returns, establishing the foundational role of correlation matrices in finance [11].

However, estimating correlation matrices from empirical financial data is problematic because of missing or asynchronous observations. Non-trading days, delisting events, reporting gaps, and corporate actions are common sources of incomplete datasets. Traditional methods often produce matrices that violate fundamental properties such as positive semidefiniteness, leading to distorted risk estimates and suboptimal financial decisions.

1.2 Problem Statement

The primary problem is the reconstruction of a valid correlation matrix from imperfect or incomplete financial datasets. A valid correlation matrix must be symmetric, positive semidefinite, and have unit diagonal entries. Common methods, such as pairwise deletion, listwise deletion, or simple imputation, often produce matrices that fail to meet these es-

sential mathematical requirements, rendering them unsuitable for financial modeling. As a result, advanced computational methods are needed to compute the nearest valid correlation matrix that best approximates the empirical data while satisfying the prescribed constraints.

Missing data introduce significant additional complexity to the nearest correlation matrix problem. According to Rubin’s framework, missingness is classified into three categories: Missing Completely at Random (MCAR), Missing at Random (MAR), and Not Missing at Random (NMAR). This thesis primarily focuses on MCAR and NMAR. Under MCAR, the absence of data points is entirely unrelated to their true values and no systematic bias is introduced. In contrast, NMAR is characterized by a nonrandom mechanism in which the probability of missingness depends directly on unobserved values, systematically inducing bias, and complicating reliable estimation [10]. Addressing NMAR conditions requires advanced reconstruction techniques that can explicitly model and correct these biases to obtain reliable correlation matrix estimates.

1.3 Research Objectives

The first objective is **Comparison of Algorithms**, which examines two methods for solving the nearest correlation matrix problem. The modified alternating projection method incorporating Dykstra’s correction ensures convergence by iteratively projecting the input matrix onto the set of symmetric positive semidefinite matrices and the set of symmetric matrices with unit diagonal entries. In contrast, the Anderson acceleration-modified alternating projection method enhances convergence speed by leveraging information from previous iterations to refine subsequent steps, making it particularly effective for large-scale matrices. Financial datasets were employed to evaluate and compare the computational efficiency, con-

vergence speed, accuracy, robustness, and scalability of both algorithms in reconstructing valid correlation matrices.

The second objective is **Missing Data Analysis**, which examines how Missing Completely at Random (MCAR) and Not Missing at Random (NMAR) patterns affect the correlation matrix estimation and algorithm performance. This thesis aims to provide recommendations for algorithm selection and data preprocessing for financial practitioners dealing with different missing data scenarios.

1.4 Motivation and Significance

This thesis is motivated by key challenges in financial modeling and risk management. In particular, two primary concerns underpin the study:

- **Financial Accuracy:** Ensuring the mathematical soundness of correlation matrices is essential, as invalid or ill-conditioned matrices can distort risk assessments. Such distortions can lead to suboptimal investment decisions and the mispricing of financial instruments.
- **Computational Efficiency:** The increasing size and complexity of financial datasets necessitate algorithms that are not only accurate but also computationally efficient and scalable. Advanced methods, such as Anderson acceleration [8], which is the focus of this thesis, offer substantial improvements in convergence speed and computational performance, making the estimation of valid correlation matrices practical for large-scale, real-world financial applications.

1.5 Scope and Limitations

This thesis focuses on algorithms that minimize the Frobenius norm to compute the nearest correlation matrix. It specifically examines two types of missing data mechanisms: Missing Completely at Random (MCAR) and Not Missing at Random (NMAR), while excluding Missing at Random (MAR) due to the need for explicit modeling of dependencies between observed and missing data. The primary applications considered are portfolio risk management and portfolio optimization within finance, although the findings may also be relevant to other fields where correlation matrices play a critical role.

The main limitations of this thesis are due to its focus on a specific class of algorithms and restricted missing data mechanisms, which do not fully capture all financial scenarios or the full complexity of real-world datasets.

1.6 Thesis Organization

This thesis is structured as follows. Chapter 2 presents a Literature Review, summarizing existing methods for computing the nearest correlation matrix and frameworks for handling missing data, with a focus on identifying research gaps in the context of financial applications. Chapter 3 establishes the Theoretical Background, providing the mathematical foundations for correlation matrix estimation and the algorithms studied. Chapter 4 details the Methodology, describing the implementation of the Modified Alternating Projections and Anderson Acceleration algorithms. Chapter 5 focuses on Missing Data Mechanisms in Statistical Analysis, exploring the definitions, implications, and financial examples of MCAR, MAR, and NMAR, and their impact on correlation matrix estimation. Chapter 6, titled Projection Methods and Missing-Data Analysis, presents the experimental setup, comparative

analysis of the algorithms, and their performance under MCAR and NMAR mechanisms, emphasizing computational speed, accuracy, robustness, and scalability. Chapter 7, titled Conclusions, summarizes the main findings, discusses implications for financial modeling, highlights limitations, and proposes directions for future research.

1.7 Core Definitions

In statistics, correlation measures the strength and direction of the linear relationship between two random variables. For two variables, a and b , the Pearson correlation coefficient is defined as follows:

$$\rho_{ab} = \frac{\text{Cov}(a, b)}{\sigma_a \sigma_b}$$

where $\text{Cov}(a, b)$ is the covariance between a and b and σ_a and σ_b are the standard deviations of a and b , respectively. The correlation coefficient ranges between -1 and 1 , with values near 1 indicating a strong positive linear association, values near -1 indicating a strong negative linear association, and values near 0 indicating little to no linear relationship. A correlation matrix extends this concept to multiple variables, capturing pairwise correlations in a structured form, as defined below.

Definition 1.7.1 (Diagonal Operator). *Let $M \in \mathbb{R}^{n \times n}$ be a square matrix. We define*

$$\text{diag}(M) := \begin{bmatrix} M_{11} \\ M_{22} \\ \vdots \\ M_{nn} \end{bmatrix} \in \mathbb{R}^n,$$

as the vector containing the diagonal elements of M .

Conversely, for a vector $v = (v_1, \dots, v_n)^\top \in \mathbb{R}^n$, we define $\text{Diag}(v) \in \mathbb{R}^{n \times n}$ as the diagonal matrix whose diagonal entries are given by the components of v , i.e.,

$$\text{Diag}(v) = \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_n \end{bmatrix}.$$

1.7.1 Correlation Matrix

A correlation matrix $X \in \mathbb{R}^{n \times n}$ satisfies the following properties:

- **Symmetry:** $X = X^\top$.
- **Positive semidefiniteness:** $X \succeq 0$, i.e., all eigenvalues of X are nonnegative.
- **Unit diagonal:** $X_{ii} = 1$ for $i = 1, \dots, n$ enforcing unit diagonal elements.

1.7.2 Nearest Correlation Matrix Problem

Given an empirical (possibly invalid) matrix $A \in \mathbb{R}^{n \times n}$, the nearest correlation matrix problem is formulated as:

$$\min_{X \in \mathcal{S}_+^n \cap \mathcal{U}^n} \|A - X\|_F,$$

where:

- $\mathcal{S}_+^n := \{Y \in \mathbb{R}^{n \times n} : Y = Y^\top, Y \succeq 0\}$ is the set of symmetric positive semidefinite matrices,

- $\mathcal{U}^n := \{Y \in \mathbb{R}^{n \times n} : Y = Y^\top, \text{diag}(Y) = e\}$ is the set of symmetric matrices with unit diagonal, where $e := (1, \dots, 1)^\top$,

and $\|\cdot\|_F$ denotes the Frobenius norm, defined for a matrix $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ as:

$$\|A\|_F^2 := \sum_{i=1}^n \sum_{j=1}^n a_{ij}^2.$$

Two specific norms are commonly employed:

- The **W -norm**, defined as

$$\|A\|_W := \|W^{1/2} A W^{1/2}\|_F,$$

where $W \in \mathbb{R}^{n \times n}$ is a symmetric positive definite matrix and $\|\cdot\|_F$ denotes the Frobenius norm. The transformation $A \mapsto W^{1/2} A W^{1/2}$ is a congruence map that preserves the inertia of A (i.e., the number of positive, negative, and zero eigenvalues), making this norm particularly useful in matrix nearness problems. The weighting matrix W can be used to emphasize specific rows and columns such as assigning more weight to more volatile or more influential stocks in a financial correlation matrix.

- The **H -norm**, defined as

$$\|A\|_H := \|H \circ A\|_F,$$

where $H \in \mathbb{R}^{n \times n}$ is a symmetric matrix of nonnegative weights and $\circ$ denotes the Hadamard (element-wise) product. This norm allows for fine-grained control over individual matrix entries: a large weight h_{ij} imposes a higher penalty for deviation between x_{ij} and a_{ij} , thereby enforcing greater accuracy in regions of high confidence. The transformation $A \mapsto H \circ A$ preserves symmetry when both A and H are symmetric.

Illustrative Example: Consider a symmetric correlation matrix $A \in \mathbb{R}^{4 \times 4}$ representing estimated relationships between four stocks:

$$A = \begin{bmatrix} 1 & 0.8 & 0.5 & 0.2 \\ 0.8 & 1 & 0.4 & 0.1 \\ 0.5 & 0.4 & 1 & 0.3 \\ 0.2 & 0.1 & 0.3 & 1 \end{bmatrix}.$$

To emphasize the relative importance of Stocks 1 and 2, a diagonal weight matrix is chosen:

$$W = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad W^{1/2} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Then, the W -norm is given by:

$$\|A\|_W = \|W^{1/2}AW^{1/2}\|_F \approx \sqrt{45.3} \approx 6.72.$$

Alternatively, to reflect entry-wise confidence levels, suppose the elementwise confidence matrix is:

$$H = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 0.5 & 0 \\ 1 & 0.5 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Then the Hadamard-weighted matrix is:

$$H \circ A = \begin{bmatrix} 1 & 1.6 & 0.5 & 0 \\ 1.6 & 1 & 0.2 & 0 \\ 0.5 & 0.2 & 1 & 0.6 \\ 0 & 0 & 0.6 & 1 \end{bmatrix}.$$

Then, the H -norm is given by:

$$\|A\|_H = \|H \circ A\|_F \approx \sqrt{10.68} \approx 3.27.$$

The choice of norm directly affects both the computational approach and the resulting matrix X . The W -norm is particularly suited to convex optimization, preserving congruence transformations and enabling efficient implementation for dense empirical datasets such as financial return matrices. By contrast, the H -norm is better suited for sparse, partial, or confidence-weighted observations due to its handling of irregular data structures. This thesis adopts the W -norm to exploit convex optimization techniques while maintaining structural consistency in the projected correlation matrices.

Definition 1.7.2 (W -weighted Inner Product and Induced Norm). *Let $W \in \mathbb{R}^{n \times n}$ be a symmetric positive definite matrix. For $A, B \in \mathbb{R}^{n \times n}$, the W -weighted inner product is defined as*

$$\langle A, B \rangle_W := \text{trace}(A^\top W B W).$$

If A and B are symmetric (i.e., $A^\top = A$, $B^\top = B$), then the W -weighted inner product simplifies to

$$\langle A, B \rangle_W = \text{trace}(A W B W),$$

and the corresponding induced norm is

$$\|A\|_W := \sqrt{\langle A, A \rangle_W} = \sqrt{\text{trace}(AWAW)} = \|W^{1/2}AW^{1/2}\|_F.$$

1.8 Summary

This chapter introduced the foundational context for the thesis, highlighting the critical role of correlation matrices in financial applications such as risk management and portfolio optimization. It outlined the challenges of reconstructing valid correlation matrices from incomplete or imperfect datasets, particularly in the presence of missing data patterns classified as MCAR and NMAR. The research objectives were defined, focusing on the comparative analysis of advanced algorithms and the assessment of missing data effects on estimation performance. The chapter also discussed the motivation and significance of the study, underscoring the need for both mathematical rigor and computational efficiency in financial modeling. The scope and limitations were clarified, and key terms and concepts were formally defined to provide a clear framework for the subsequent chapters. This sets the stage for a detailed exploration of the existing methods and the development of robust solutions in the chapters that follow.

CHAPTER 2

LITERATURE REVIEW

2.1 Methods for Computing the Nearest Correlation Matrix

The nearest correlation matrix (NCM) problem modifies an empirically derived matrix to satisfy the properties of a valid correlation matrix: symmetry, unit diagonal, and positive semidefiniteness. Such violations frequently arise due to asynchronous data acquisition, missing data, stress-testing changes, or numerical inaccuracies.

Higham [7] introduced the first broadly embraced method with guaranteed convergence: the modified alternating projections algorithm. This iterative algorithm alternately projects the input matrix onto two sets: the set of symmetric positive semidefinite matrices (by eigendecomposition, by resetting any negative eigenvalue to zero) and the set of symmetric matrices with a unit diagonal. The algorithm converges to the nearest correlation matrix in Frobenius norm. Its convergence rate, nonetheless, is linear and hence can be slow for large or ill-conditioned matrices. Subsequently, a Newton-type method with quadratic convergence was proposed by Qi and Sun [9], and later refined by Borsdorf and Higham [6] to enhance its practical performance.

Despite these advances, the alternating projections method remains widely used in practice due to its simplicity, numerical robustness, and the availability of efficient implementations across popular platforms such as MATLAB, Python, R, SAS, and Julia. Its ease of deployment and reliable performance make it a competitive choice in financial and statistical applications, even when faster algorithms are available.

2.1.1 Modified Alternating Projections

To rectify the slow convergence of basic alternating projections, Higham [7] incorporated Dykstra’s correction originally devised by Boyle and Dykstra [12]. This adjustment applies correction terms at each projection step to correct for accumulated errors from previous projections, reduce oscillations, and ensure a more stable convergence. Subsequent research has explored and applied the modified alternating projection method in finance-motivated settings, highlighting its relevance for generating valid correlation matrices used in risk assessment and portfolio modeling.

2.1.2 Anderson Acceleration

Anderson acceleration, originally developed by Anderson [13], significantly improves the efficiency of fixed-point iterations by utilizing information from previous steps to enhance subsequent updates. Higham and Strabić [8] applied this method to the nearest correlation matrix (NCM) problem and demonstrated that it can reduce the number of iterations by up to half in typical scenarios and by factors of three or more in more challenging cases, such as when fixed elements (i.e., constraints that enforce specific matrix entries to remain unchanged) or eigenvalue bounds are imposed. Anderson acceleration delivers an improved practical performance for large-scale financial computations with minimal additional computational overhead.

2.1.3 Missing Data Mechanisms

The classification of missing data mechanisms was first formalized by Rubin [10], who introduced a probabilistic framework by treating missing data indicators as random variables with associated distributions. This work marked a paradigm shift, moving from ad hoc methods to a model-based approach, laying the groundwork for modern missing data theory. Little and Rubin built upon this foundation, refining terminology and notation, and introducing the missing data indicator R and parameter ϕ to describe the relationship between missingness and data [18]. Their seminal text, *Statistical Analysis with Missing Data*, clarified that these mechanisms represent mathematical relationships rather than causal explanations. Enders further emphasized this distinction, noting that missing data mechanisms describe probabilistic dependencies, guiding researchers in method selection [19]. In the context of financial time-series, Burger et al. highlight the prevalence of missing data due to market closures or stale data, underscoring the practical importance of these classifications in financial modeling [20].

2.1.4 Literature Gap Analysis

Existing work on nearest correlation matrix (NCM) methods has not evaluated algorithmic performance across varying degrees of missing data under MCAR and NMAR mechanisms. Higham and Strabić examined the computational behavior of alternating projections and related acceleration schemes on fully observed matrices [8]. Chen and Li analyzed how different missingness patterns affect statistical estimators without assessing matrix reconstruction techniques [14]. Burger et al. developed procedures for deriving correlation estimates from incomplete financial time-series, illustrating practical challenges posed by

real-world data gaps [20]. Yet no study has benchmarked modified alternating projections against Anderson acceleration under both MCAR and NMAR mechanisms. This omission restricts the applicability of NCM algorithms to financial datasets, where nonrandom missingness is common, and impedes the creation of resilient computational tools.

2.2 Research Gap Identification

This thesis presents a systematic comparison of modified alternating projections and Anderson acceleration under Missing Completely at Random (MCAR) and Not Missing at Random (NMAR) data mechanisms. The evaluation of computational efficiency and reconstruction accuracy demonstrates that Anderson acceleration consistently outperforms the modified alternating projections method in terms of convergence speed and robustness, particularly in NMAR conditions. These findings provide empirical insights and practical guidance for financial modeling, contributing to the improved reliability of correlation matrix estimation under realistic data conditions.

CHAPTER 3

THEORETICAL BACKGROUND

This chapter establishes the mathematical foundation for computing nearest correlation matrices (NCM). We begin by formally defining the problem and characterizing optimal solutions through convex analysis, setting the stage for numerical algorithms developed in Chapter 4.

3.1 Definition and Significance of Correlation Matrices

A correlation matrix is a symmetric, positive semidefinite matrix with unit diagonal entries. Formally, a matrix $X \in \mathbb{R}^{n \times n}$ is a correlation matrix if it satisfies:

- $X = X^\top$, ensuring symmetry;
- $X \succeq 0$, indicating positive semidefiniteness;
- $X_{ii} = 1$ for $i = 1, \dots, n$ enforcing unit diagonal elements.

These properties arise naturally in statistics, where correlation matrices represent the correlation coefficients between random variables. Beyond statistics, correlation matrices play a significant role in numerical linear algebra, including applications in preconditioning linear systems and analyzing errors in Jacobi methods for symmetric eigenvalue problems [1].

In financial applications, correlation matrices are particularly relevant for modeling relationships between asset returns. Sample correlation matrices, constructed from vectors of

stock returns, are often used for predictive purposes in portfolio optimization and risk management. However, these empirical matrices may not be positive semidefinite or have unit diagonal entries due to incomplete or inconsistent data sets, such as missing stock return data on certain days [7] or judgmental modifications (e.g., during stress testing or expert-driven adjustments). This necessitates computing the nearest valid correlation matrix to ensure the reliability of subsequent analyses. Such matrices are typically dense, with dimensions often in the thousands, and are characterized by low rank when derived from limited observational data.

3.1.1 Formulation of the Nearest Correlation Matrix Problem

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, the nearest correlation matrix problem seeks to find a correlation matrix X that minimizes the distance to A . Mathematically, this is expressed as:

$$\gamma(A) := \min_{X \in \mathcal{S}_+^n \cap \mathcal{U}^n} \|A - X\|_W, \quad (3.1)$$

where:

- $\mathcal{S}_+^n = \{Y \in \mathbb{R}^{n \times n} : Y = Y^\top, Y \succeq 0\}$ is the set of symmetric positive semidefinite matrices,
- $\mathcal{U}^n = \{Y \in \mathbb{R}^{n \times n} : Y = Y^\top, \text{diag}(Y) = e\}$ is the set of symmetric matrices with unit diagonal.

The intersection $\mathcal{S}_+^n \cap \mathcal{U}^n$ defines the set of correlation matrices. The objective is to compute both $\gamma(A)$, the minimal distance, and the matrix X that achieves this minimum.

The distance $\|A - X\|_W$ is measured using weighted Frobenius norms, which allow flexibility in assigning varying levels of importance or confidence to different entries of the input matrix A .

3.2 Characterization of the Solution via Convex Analysis

To characterize the solution X to the nearest correlation matrix problem, we employ the framework of convex analysis, which provides a principled approach to understanding the optimization landscape. The solution $X \in \mathcal{S}_+^n \cap \mathcal{U}^n$ to problem (3.1) is characterized by a standard result from convex projection theory [4]. A matrix $X \in \mathcal{S}_+^n \cap \mathcal{U}^n$ is a minimizer of $\|A - X\|_W$ if and only if

$$\langle Z - X, A - X \rangle_W \leq 0, \text{ for all } Z \in \mathcal{S}_+^n \cap \mathcal{U}^n. \quad (3.2)$$

This condition can be equivalently expressed using the concept of the normal cone to a convex set. For a closed convex set $K \subseteq \mathbb{R}^{n \times n}$, the normal cone at a point $B \in K$ is defined as

$$\partial K(B) := \{Y \in \mathbb{R}^{n \times n} : \langle Z - B, Y \rangle_W \leq 0, \text{ for all } Z \in K\}, \quad (3.3)$$

or, equivalently,

$$\partial K(B) := \left\{ Y \in \mathbb{R}^{n \times n} : \langle Y, B \rangle_W = \sup_{Z \in K} \langle Y, Z \rangle_W \right\}. \quad (3.4)$$

Applying this to $K = \mathcal{S}_+^n \cap \mathcal{U}^n$, the optimal solution X satisfies

$$A - X \in \partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X). \quad (3.5)$$

Since $\text{relint}(\mathcal{S}_+^n) = \mathcal{S}_{++}^n$ and $\text{relint}(\mathcal{U}^n) = \mathcal{U}^n$, we have $\text{relint}(\mathcal{S}_+^n) \cap \text{relint}(\mathcal{U}^n) \neq \emptyset$ (in fact $I \in \mathcal{S}_{++}^n \cap \mathcal{U}^n$). By the normal cone sum rule [2, Cor. 23.8.1], the normal cone to their intersection decomposes as

$$\partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X) = \partial\mathcal{S}_+^n(X) + \partial\mathcal{U}^n(X). \quad (3.6)$$

This decomposition allows the optimality condition (3.5) to be verified by analyzing $\partial\mathcal{S}_+^n(X)$ and $\partial\mathcal{U}^n(X)$ separately, simplifying the characterization of the solution.

3.3 Normal Cones of Feasible Sets

This section derives the normal cones to the feasible sets $\mathcal{U}^n$ and $\mathcal{S}_+^n$, which define the set of valid correlation matrices $\mathcal{S}_+^n \cap \mathcal{U}^n$ in the nearest correlation matrix problem (3.1). These characterizations are essential for analyzing the optimality condition in (3.5), deriving the solution structure in Section 3.4, and developing projection algorithms in Section 3.6.

3.3.1 Normal Cone of the Set $\mathcal{U}^n$

The normal cone to the set $\mathcal{U}^n$, which comprises of symmetric matrices with unit diagonal, is described by Higham [7, Lem. 2.1].

Lemma 3.3.1. *For $A \in \mathcal{U}^n$, the normal cone of the set $\mathcal{U}^n$ at A , under the W -weighted inner product, is given by*

$$\partial\mathcal{U}^n(A) = \{W^{-1} \text{Diag}(\theta_i)W^{-1} : \theta_i \in \mathbb{R}^n\}.$$

Proof. We have:

$$\partial\mathcal{U}^n(A) = \left\{ Y = Y^\top \in \mathbb{R}^{n \times n} : \langle Y, A \rangle_W = \sup_{Z \in \mathcal{U}^n} \langle Y, Z \rangle_W \right\}.$$

The inner product is defined as $\langle Y, Z \rangle_W := \text{trace}(YWZW) = \text{trace}(\tilde{Y}Z)$, where $\tilde{Y} := WYW$.

The constraint can be written:

$$\sum_{i,j} \tilde{y}_{ij} a_{ij} = \sup_{Z \in \mathcal{U}^n} \sum_{i,j} \tilde{y}_{ij} z_{ij}.$$

Since $Z \in \mathcal{U}^n$, the diagonal entries $z_{ii} = 1$, while the off-diagonal entries z_{ij} (for $i \neq j$) are arbitrary. If $\tilde{Y}$ has a nonzero off-diagonal entry $\tilde{y}_{ij} \neq 0$ for $i \neq j$, we can choose $z_{ij} \in \mathbb{R}$ with the same sign as $\tilde{y}_{ij}$ and arbitrarily large magnitude, making the right-hand side unbounded. This contradicts the equality, so $\tilde{y}_{ij} = 0$ for all $i \neq j$. Thus, $\tilde{Y}$ must be diagonal, and we have:

$$\tilde{Y} = \text{Diag}(\theta_i), \quad \text{and hence } Y = W^{-1} \text{Diag}(\theta_i) W^{-1},$$

where $\theta_i \in \mathbb{R}$ are arbitrary real numbers. □

3.3.2 Normal Cone of the Set $\mathcal{S}_+^n$

The normal cone to the set $\mathcal{S}_+^n$, comprising of symmetric positive semidefinite matrices, is characterized by two key results by Higham [7, Lem 2.2, Cor 2.3]. The following lemma provides a variational description of $\partial\mathcal{S}_+^n(A)$ under the weighted inner product.

Lemma 3.3.2. *For $A \in \mathcal{S}_+^n$, the normal cone to $\mathcal{S}_+^n$ at A , under the W -weighted inner product, is given by*

$$\partial\mathcal{S}_+^n(A) = \{Y = Y^\top \in \mathbb{R}^{n \times n} : \langle Y, A \rangle_W = 0, Y \preceq 0\},$$

where $\langle Y, A \rangle_W := \text{trace}(YWA W)$, and $Y \preceq 0$ denotes that Y is symmetric and negative semidefinite.

Proof. By definition of the normal cone, we have

$$\partial\mathcal{S}_+^n(A) = \left\{ Y = Y^\top \in \mathbb{R}^{n \times n} : \langle Y, A \rangle_W = \sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W \right\}.$$

Suppose $Y = Y^\top \in \mathbb{R}^{n \times n}$. Let $Z \in \mathcal{S}_+^n$ admit a spectral decomposition

$$Z := Q\Lambda Q^\top,$$

where $Q \in \mathbb{R}^{n \times n}$ is orthogonal and $\Lambda = \text{Diag}(\lambda_1, \dots, \lambda_n) \succeq 0$. Define

$$C := Q^\top W Y W Q.$$

Then:

$$\begin{aligned} \langle Y, Z \rangle_W &= \text{trace}(Y W Z W), \\ &= \text{trace}(Y W Q \Lambda Q^\top W), \\ &= \text{trace}(Q^\top W Y W Q \Lambda), \end{aligned}$$

and since Λ is diagonal,

$$\langle Y, Z \rangle_W = \langle C, \Lambda \rangle.$$

Hence,

$$\begin{aligned} \sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W &= \sup_{\Lambda \succeq 0, Q^\top Q = I} \langle C, \Lambda \rangle \\ &= \sup_{\Lambda \succeq 0, Q^\top Q = I} \sum_i \lambda_i c_{ii}. \end{aligned}$$

If $Y \preceq 0$, then $C = Q^\top W Y W Q \preceq 0$, which implies $c_{ii} \leq 0$ for all i . Since $\lambda_i \geq 0$ for all $Z \in \mathcal{S}_+^n$, we have $\sum_i \lambda_i c_{ii} \leq 0$, and the supremum is attained at $\Lambda = 0$. Thus,

$$\sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W = 0.$$

If $Y \not\preceq 0$, then there exists some $c_{ii} > 0$, and we can choose $\lambda_i \rightarrow \infty$, giving

$$\sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W = +\infty.$$

Therefore,

$$\sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W = \begin{cases} 0, & \text{if } Y \preceq 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Equality holds in the supremum condition for such Y satisfying $\langle Y, A \rangle_W = 0$ and $Y \preceq 0$.

Conversely, assume $Y = Y^\top \in \mathbb{R}^{n \times n}$, $Y \preceq 0$, and $\langle Y, A \rangle_W = 0$. From the above, we have $\sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W = 0$, and thus,

$$\langle Y, A \rangle_W = \sup_{Z \in \mathcal{S}_+^n} \langle Y, Z \rangle_W,$$

which implies by definition that

$$Y \in \partial \mathcal{S}_+^n(A).$$

Combining both inclusions, we conclude that

$$\partial\mathcal{S}_+^n(A) = \{Y = Y^\top \in \mathbb{R}^{n \times n} : \langle Y, A \rangle_W = 0, Y \preceq 0\}.$$

□

A more specific and computationally practical form of $\partial\mathcal{S}_+^n(A)$ can be obtained by exploring the null space of A .

Corollary 3.3.2.1. *For $A \in \mathcal{S}_+^n$, the elements of the normal cone can be expressed as*

$$\partial\mathcal{S}_+^n(A) = \{Y : WYW = -VDV^\top, D = \text{Diag}(d_i) \succeq 0\},$$

where $V \in \mathbb{R}^{n \times p}$ has orthonormal columns spanning $\text{null}(A)$.

Proof. Let $A \in \mathcal{S}_+^n$ have the spectral decomposition

$$A = Q\Lambda Q^\top, \quad \Lambda = \text{Diag}(\lambda_1, \dots, \lambda_n), \text{ with } \lambda_1 \geq \dots \geq \lambda_{n-p} > 0 = \lambda_{n-p+1} = \dots = \lambda_n.$$

Write $Q := [Q_1, Q_2]$, where $Q_1 \in \mathbb{R}^{n \times (n-p)}$ corresponds to the positive eigenvalues and the columns of $Q_2 \in \mathbb{R}^{n \times p}$ span $\text{null}(A)$. From Lemma 3.3.2, for $Y \in \partial\mathcal{S}_+^n(A)$

$$0 = \langle Y, A \rangle_W = \text{trace}(AWYW).$$

Using the spectral decomposition, we compute:

$$\text{trace}(AWYW) = \text{trace}(Q_1\Lambda_1Q_1^\top WYW) = \text{trace}(\Lambda_1Q_1^\top WYWQ_1),$$

where $\Lambda_1 = \text{Diag}(\lambda_1, \dots, \lambda_{n-p}) \succ 0$. Since $Y \preceq 0$, this implies

$$\text{diag}(Q_1^\top WYW Q_1) = 0.$$

Consider the block matrix representation of $Q^\top(WYW)Q$ to analyze the structure of WYW :

$$Q^\top(WYW)Q = \begin{bmatrix} Q_1^\top(WYW)Q_1 & Q_1^\top(WYW)Q_2 \\ Q_2^\top(WYW)Q_1 & Q_2^\top(WYW)Q_2 \end{bmatrix} = \begin{bmatrix} G & H \\ H^\top & M \end{bmatrix} \preceq 0,$$

where $G := Q_1^\top WYW Q_1$, $H := Q_1^\top WYW Q_2$, and $M := Q_2^\top WYW Q_2$. Since $\text{Diag}(G) = 0$, it follows that $G = 0$. As the matrix is negative semidefinite, it follows that $H = 0$, as nonzero H would violate negative semidefiniteness. Furthermore, $M \preceq 0$. Thus:

$$WYW = Q \begin{bmatrix} 0 & 0 \\ 0 & M \end{bmatrix} Q^\top = Q_2 M Q_2^\top,$$

where, $WYW = Q_2 M Q_2^\top = -VDV^\top$. We have used the spectral decomposition of M to produce $V \in \mathbb{R}^{n \times p}$ with orthonormal columns spanning $\text{null}(A)$ and $D = \text{Diag}(M) \succeq 0 \in \mathbb{R}^{p \times p}$ diagonal and positive semidefinite. $\square$

3.4 Solution Characterization

The characterizations of $\partial\mathcal{U}^n(X)$ and $\partial\mathcal{S}_+^n(X)$ provide the foundation for the Theorem 3.4.1 and Theorem 3.5.1, which describe the structure of the solution X to the nearest correlation matrix problem and its properties, particularly in financial applications [7, Thm. 2.4].

Theorem 3.4.1. *The correlation matrix X solves (3.1) if and only if*

$$X = A + W^{-1}(VDV^\top + \text{Diag}(\theta_i))W^{-1}, \quad (3.7)$$

where $V \in \mathbb{R}^{n \times p}$ has orthonormal columns spanning $\text{null}(X)$, $D = \text{Diag}(d_i) \succeq 0$, and $\theta_i \in \mathbb{R}$ are arbitrary.

Proof. ($\Rightarrow$) Assume X solves (3.1). Then $X \in \mathcal{S}_+^n \cap \mathcal{U}^n$, and

$$A - X \in \partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X) = \partial\mathcal{S}_+^n(X) + \partial\mathcal{U}^n(X).$$

By Corollary 3.3.2.1,

$$\begin{aligned} \partial\mathcal{S}_+^n(X) &= \{Y = Y^\top : WYW = -VDV^\top, V^\top V = I, \\ &\quad \text{col}(V) = \text{null}(X), \text{Diag}(d_i) \succeq 0\}. \end{aligned}$$

By Lemma 3.3.1,

$$\partial\mathcal{U}^n(X) = \{W^{-1} \text{Diag}(\theta_i)W^{-1} : \theta_i \in \mathbb{R}\}.$$

Thus,

$$A - X = W^{-1}(-VDV^\top)W^{-1} + W^{-1} \text{Diag}(\theta_i)W^{-1},$$

yielding

$$X = A + W^{-1}(VDV^\top - \text{Diag}(\theta_i))W^{-1}.$$

Since θ_i are arbitrary, we may reverse their sign to absorb the minus term, leaving the representation of X unchanged. Hence, we can equivalently write

$$X = A + W^{-1}(VDV^\top + \text{Diag}(\theta_i))W^{-1}.$$

($\Leftarrow$) Assume

$$X = A + W^{-1}(VDV^\top + \text{Diag}(\theta_i))W^{-1},$$

with $V \in \mathbb{R}^{n \times p}$, $V^\top V = I$, $\text{col}(V) = \text{null}(X)$, $D = \text{Diag}(d_i) \succeq 0$, and $\theta_i \in \mathbb{R}$. Then

$$W^{-1}VDV^\top W^{-1} \succeq 0,$$

since $D \succeq 0$ and conjugation by W^{-1} preserves positive semidefiniteness.

By Corollary 3.3.2.1,

$$W^{-1}VDV^\top W^{-1} \in \partial \mathcal{S}_+^n(X),$$

and by Lemma 3.3.1,

$$W^{-1} \text{Diag}(\theta_i) W^{-1} \in \partial \mathcal{U}^n(X).$$

Hence,

$$A - X = W^{-1}(VDV^\top + \text{Diag}(\theta_i))W^{-1} \in \partial \mathcal{S}_+^n(X) + \partial \mathcal{U}^n(X) = \partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X).$$

Since

$$A - X \in \partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X),$$

X is optimal, and strict convexity of $\|\cdot\|_W$ ensures uniqueness of the minimizer.

□

The implications of Theorem 3.4.1 are significant. The term $VDV^\top$ ensures that X remains positive semidefinite, as it adjusts A by a negative semidefinite component in the direction of the null space of X . The diagonal term $\text{Diag}(\theta_i)$ allows for adjustments to enforce the unit diagonal constraint of $\mathcal{U}^n$. When W is diagonal, the solution X is generally singular (i.e., not positive definite), as the presence of a nontrivial null space ($V \neq 0$) introduces

zero eigenvalues. The only case where X may be nonsingular is when $V = 0$, implying $\text{null}(X) = \{0\}$, in which case:

$$X = A + W^{-1} \text{Diag}(\theta_i) W^{-1}, \quad (3.8)$$

indicating that X is obtained by adjusting the diagonal elements of A to be 1. Additionally, (3.7) implies a necessary condition for optimality:

$$XW(X - A)W = X \text{Diag}(\theta_i). \quad (3.9)$$

The parameters θ_i are arbitrary here but are constrained in Section 3.5 (e.g., $\theta_i \leq 0$ when $a_{ii} \geq 1$) to ensure specific solution properties.

3.5 Constraints and Eigenvalue Analysis

Higham [7, Thm. 2.5] discusses the properties for financial applications. It provides critical insights into the properties of the solution X when the input matrix $A = A^\top$ has diagonal elements $a_{ii} \geq 1$ and W is diagonal. Under these conditions, the theorem states:

- The adjustments θ_i in (3.7) are nonpositive, i.e., $\theta_i \leq 0$ for all i .
- If A has t nonpositive eigenvalues, then the nearest correlation matrix X has at least t zero eigenvalues, implying that the dimension of the null space of X , denoted p , satisfies $p \geq t$.

Theorem 3.5.1. *Let $A = A^\top$ have diagonal elements $a_{ii} \geq 1$, and let W be diagonal. Then, in the representation of the optimal solution from Theorem 3.4.1, we must have $\theta_i \leq 0$ for*

all i . Moreover, if A has t nonpositive eigenvalues, then the nearest correlation matrix has at least t zero eigenvalues.

Proof. Since A has diagonal elements $a_{ii} \geq 1$, the perturbation $W^{-1}(VDV^\top)W^{-1} \succeq 0$ in the solution

$$X = A + W^{-1}(VDV^\top + \text{Diag}(\theta_i))W^{-1} \quad (3.10)$$

ensures that the diagonal elements of $A + W^{-1}(VDV^\top)W^{-1}$ are at least 1. To satisfy the unit diagonal constraint $x_{ii} = 1$, it is required that the diagonal correction $\text{Diag}(\theta_i)$ satisfies $\theta_i \leq 0$ for all i .

Next, we examine the eigenvalue structure. The correction term

$$W^{-1}(VDV^\top)W^{-1}$$

adjusts the eigenvalues of

$$A + W^{-1} \text{Diag}(\theta_i)W^{-1}$$

to ensure $X \succeq 0$. The matrix $A + W^{-1} \text{Diag}(\theta_i)W^{-1}$ may have t or more nonpositive eigenvalues, depending on θ_i .

The term $VDV^\top$ is positive semidefinite and has rank at most p , where $p = \dim(\text{null}(X))$. By a standard result on eigenvalue shifts under symmetric low-rank perturbations [15, Thm. 4.3.6], at least p nonpositive eigenvalues of

$$A + W^{-1} \text{Diag}(\theta_i)W^{-1}$$

can be shifted to nonnegative values only if $p \geq t$. This implies that the null-space dimension of X is sufficient to correct the t nonpositive eigenvalues; that is, since $p \geq t$, the rank of

the correction term accommodates the required shifts. Consequently, the solution X must have at least t zero eigenvalues.

□

For example, consider the invalid correlation matrix

$$A = \begin{pmatrix} 1 & 1.01 \\ 1.01 & 1 \end{pmatrix},$$

which has eigenvalues 2.01 and -0.01 . By Theorem 3.5.1, the nearest correlation matrix X must have at least one zero eigenvalue and satisfy $\theta_i \leq 0$ for the diagonal correction. Thus,

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

whose eigenvalues are 2 and 0, unit diagonal entries hold, and $X \succeq 0$, so $X \in \mathcal{S}_2 \cap \mathcal{U}_2$.

3.6 Projection Operators for Correlation Matrices

The objective is to project a given symmetric matrix onto the intersection of two constraint sets: the set $\mathcal{S}_+^n$ of symmetric positive semidefinite matrices, and the set $\mathcal{U}^n$ of matrices with unit diagonal entries. To approach this, we first analyze the projection onto each set individually. We begin with the projection onto $\mathcal{S}_+^n$, denoted by $\mathcal{P}_{\mathcal{S}_+^n}$, and subsequently consider the projection onto $\mathcal{U}^n$, denoted by $\mathcal{P}_{\mathcal{U}^n}$.

The projection operator $\mathcal{P}_{\mathcal{S}_+^n}(A)$, which projects a symmetric matrix A onto the cone of symmetric positive semidefinite matrices $\mathcal{S}_+^n$ under the weighted Frobenius norm, plays a central role in solving the nearest correlation matrix problem (3.1). This operation is a

fundamental step in iterative correction algorithms for restoring invalid empirical correlation matrices, as formalized in [7, Thm. 3.1].

Theorem 3.6.1 (Projection onto $\mathcal{S}_+^n$ in the W -norm). *For the W -norm, the projection of a symmetric matrix $A \in \mathbb{R}^{n \times n}$ onto $\mathcal{S}_+^n$ is given by:*

$$\mathcal{P}_{\mathcal{S}_+^n}(A) := W^{-1/2} \left((W^{1/2} A W^{1/2})_+ \right) W^{-1/2}.$$

Moreover,

$$\text{diag} \left(\mathcal{P}_{\mathcal{S}_+^n}(A) \right) \geq \text{diag}(A).$$

Proof. We need to verify that the proposed projection $X := \mathcal{P}_{\mathcal{S}_+^n}(A)$ satisfies the optimality condition from Lemma 3.3.2:

$$A - X \in \partial \mathcal{S}_+^n(X), \quad \text{i.e., } A - X \preceq 0, \quad \text{trace}((A - X)W X W) = 0.$$

To confirm $A - X \preceq 0$, we transform the residual using the weighted norm structure. Compute:

$$\begin{aligned} A - X &= A - W^{-1/2} (W^{1/2} A W^{1/2})_+ W^{-1/2} \\ &= W^{-1/2} \left(W^{1/2} A W^{1/2} - (W^{1/2} A W^{1/2})_+ \right) W^{-1/2} \\ &= W^{-1/2} (W^{1/2} A W^{1/2})_- W^{-1/2} \preceq 0, \end{aligned}$$

since $(W^{1/2}AW^{1/2})_-$ contains only nonpositive eigenvalues. Next, we verify the orthogonality condition:

$$\begin{aligned}(A - X)WXW &= W^{-1/2} (W^{1/2}AW^{1/2})_- W^{-1/2} \cdot W^{1/2} (W^{1/2}AW^{1/2})_+ W^{1/2} \\ &= W^{-1/2} (W^{1/2}AW^{1/2})_- (W^{1/2}AW^{1/2})_+ W^{1/2} = 0.\end{aligned}$$

The matrices $(W^{1/2}AW^{1/2})_+$ and $(W^{1/2}AW^{1/2})_-$ share the same eigenvectors but have disjoint eigenvalue supports (nonnegative and negative), so their product is zero. Thus:

$$\text{trace}((A - X)WXW) = 0.$$

Finally, for the diagonal condition:

$$(W^{1/2}AW^{1/2})_+ - W^{1/2}AW^{1/2} = - (W^{1/2}AW^{1/2})_- \succeq 0.$$

Pre- and post-multiplying by $W^{-1/2}$ creates a congruence transformation, which preserves the positive semidefiniteness of a matrix and, consequently, any matrix inequality. Applying this transformation and then taking the diagonal yields the desired result, since the diagonal entries of any positive semidefinite matrix are nonnegative. Specifically, we obtain:

$$\text{diag}(X) \geq \text{diag}(A),$$

which concludes the proof. □

Remark 3.6.1. *The projection onto $\mathcal{S}_+^n$ requires an eigenvalue decomposition, with a computational complexity of $\mathcal{O}(n^3)$ due to the cost of computing the full spectral decomposition of an $n \times n$ matrix [7]. This complexity is significant for large-scale financial applications, where efficient algorithms are critical.*

The projection operator $\mathcal{P}_{\mathcal{U}^n}(A)$, which maps a symmetric matrix A onto the set $\mathcal{U}^n$ of matrices with unit diagonal entries using the weighted Frobenius norm, makes sure that the diagonal values are all equal to one. This step works together with the projection onto the set $\mathcal{S}_+^n$ of positive semidefinite matrices [7, Thm. 3.2].

The Hadamard product $\circ$ denotes elementwise multiplication. For $W \in \mathbb{R}^{n \times n}$ with $W^{-1} = (\omega_{ij})$, the matrix $W^{-1} \circ W^{-1}$ consists of the squared entries of the inverse:

$$(W^{-1} \circ W^{-1})_{ij} := \omega_{ij}^2.$$

This operation preserves the dimension of W^{-1} while applying entrywise squaring.

Theorem 3.6.2 (Projection onto $\mathcal{U}^n$ in the W -norm). *For the W -norm, the projection of a symmetric matrix $A \in \mathbb{R}^{n \times n}$ onto $\mathcal{U}^n$ is given by*

$$\mathcal{P}_{\mathcal{U}^n}(A) := A - W^{-1} \text{Diag}(\theta_i) W^{-1},$$

where $\theta = [\theta_1, \dots, \theta_n]^\top \in \mathbb{R}^n$ solves the linear system:

$$(W^{-1} \circ W^{-1})\theta = \text{diag}(A - I). \quad (3.11)$$

Proof. Let $X = \mathcal{P}_{\mathcal{U}^n}(A)$. From Lemma 3.3.1, the optimality condition is

$$A - X \in \partial \mathcal{U}^n(X),$$

implying

$$A - X = W^{-1} \text{Diag}(\theta_i) W^{-1}$$

for some vector $\theta \in \mathbb{R}^n$.

To enforce $x_{ii} = 1$, we compute the diagonal elements of X . Writing $W^{-1} = (\omega_{ij})$, we have:

$$x_{ii} = a_{ii} - \sum_{j=1}^n \omega_{ij}^2 \theta_j.$$

Setting $x_{ii} = 1$ for all i , we obtain:

$$\sum_{j=1}^n \omega_{ij}^2 \theta_j = a_{ii} - 1.$$

This forms the linear system:

$$(W^{-1} \circ W^{-1})\theta = \text{diag}(A - I).$$

Since $W \succ 0$, we have $W^{-1} \succ 0$, so $W^{-1} \circ W^{-1} \succ 0$. Thus, the linear system (3.11) has a unique solution. $\square$

For example, consider the symmetric matrix

$$A = \begin{pmatrix} 1.2 & 0.9 \\ 0.9 & 0.8 \end{pmatrix},$$

with non-unit diagonals. For $W = I$, solving (3.11) yields $\theta_1 = 0.2$, $\theta_2 = -0.2$, so

$$\mathcal{P}_{\mathcal{U}^n}(A) = A - \text{Diag}(0.2, -0.2) = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}.$$

This projection ensures unit diagonals, aligning with the Solution Characterization Theorem 3.4.1 and supporting financial modeling [7].

Remark 3.6.2. *This projection enforces the unit diagonal, essential for correlation matrices, and is computationally efficient for diagonal W , requiring $\mathcal{O}(n)$ operations due to the*

diagonal structure of the linear system (3.11). This efficiency enhances iterative algorithms for the NCM problem [7].

CHAPTER 4

METHODOLOGY

4.1 Alternating Projections Method

In prior chapters, we presented the Nearest Correlation Matrix (NCM) problem, which seeks to identify a matrix X within the intersection of two convex sets: the positive semidefinite cone $\mathcal{S}_+^n$, comprising all symmetric positive semidefinite matrices in $\mathbb{R}^{n \times n}$, and the set $\mathcal{U}^n$, encompassing symmetric matrices with unit diagonal entries. The objective is to minimize the weighted Frobenius norm distance, defined with respect to a weight matrix W , between a given symmetric matrix A and a valid correlation matrix X . As outlined in Chapter 3, this problem is significant in applications such as finance, where empirical matrices A may fail to satisfy the properties of a valid correlation matrix.

To address this problem, we employ the alternating projections method, an iterative algorithm that alternately projects an estimate onto the sets $\mathcal{S}_+^n$ and $\mathcal{U}^n$ to converge to a point in their intersection. While von Neumann's classical method ensures convergence for projections onto subspaces [17], our analysis in Chapter 3 indicates that $\mathcal{S}_+^n$ and $\mathcal{U}^n$ are convex sets but not subspaces. However, when dealing with closed convex sets, simple iterative projection can converge to non-optimal points. To address this, a modified alternating projections method incorporating Dykstra's correction to ensure convergence to the optimal solution in $\mathcal{S}_+^n$ and $\mathcal{U}^n$ is used.

4.1.1 Modified Alternating Projections

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, this algorithm computes the nearest correlation matrix Y to A by MAP. It requires a convergence tolerance tol . Although the nearest correlation matrix problem is formulated and analyzed using the W -norm throughout Chapters 3, the convergence criterion in this algorithm and all subsequent numerical evaluations use the Frobenius norm. This choice is made for practical interpretability and alignment with existing literature.

Algorithm 1 Modified Alternating Projections (MAP)

- 1: **Input:** Symmetric matrix $A \in \mathbb{R}^{n \times n}$, tolerance $\text{tol} > 0$
 - 2: **Output:** Approximate correlation matrix $Y_k \in \mathbb{R}^{n \times n}$
 - 3: **:** $\Delta S_0 \leftarrow 0, Y_0 \leftarrow A$
 - 4: **for** $k = 1, 2, \dots$ **do**
 - 5: $R_k \leftarrow Y_{k-1} - \Delta S_{k-1}$ ▷ Dykstra correction term
 - 6: $X_k \leftarrow \mathcal{P}_{S_+^n}(R_k)$ ▷ Project onto PSD cone
 - 7: $\Delta S_k \leftarrow X_k - R_k$ ▷ Update correction term
 - 8: $Y_k \leftarrow \mathcal{P}_{\mathcal{U}^n}(X_k)$ ▷ Project onto unit-diagonal set
 - 9: **if** $\|Y_k - X_k\|_F \leq \text{tol} \cdot \|Y_k\|_F$ **then**
 - 10: **Return** Y_k
-

The termination criterion in line 9 is a simplified form of the general relative stopping condition, expressed in the Frobenius norm:

$$\max \left\{ \frac{\|X_k - X_{k-1}\|_F}{\|X_k\|_F}, \frac{\|Y_k - Y_{k-1}\|_F}{\|Y_k\|_F}, \frac{\|Y_k - X_k\|_F}{\|Y_k\|_F} \right\} \leq \text{tol.}$$

This condition monitors the relative change in both iterates and projections, ensuring convergence is not declared prematurely due to local stagnation in only one of the variables. In practice, however, simplified criteria such as monitoring only $\|X_k - X_{k-1}\|_F$ or $\|Y_k - X_k\|_F$ are often sufficient and more efficient, particularly when the iterates remain well-scaled.

4.1.2 Convergence of Algorithm 1

The convergence of Dykstra's algorithm is well-established by general results in convex optimization. Boyle and Dykstra [12, Thm. 2] and Han [5, Thm. 4.7] demonstrate that for any two closed convex sets, such as the positive semidefinite cone $\mathcal{S}_+^n$ and the unit-diagonal set $\mathcal{U}^n$, the sequences $\{X_k\}$ and $\{Y_k\}$ generated by Algorithm 1 converge to the nearest correlation matrix $X \in \mathcal{S}_+^n \cap \mathcal{U}^n$ that minimizes the weighted Frobenius norm $\|A - X\|_F$. Specifically, as $k \rightarrow \infty$, we have $X_k \rightarrow X$ and $Y_k \rightarrow X$, where X satisfies the optimality condition

$$A - X \in \partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X),$$

with $\partial(\mathcal{S}_+^n \cap \mathcal{U}^n)(X)$ denoting the normal cone to the intersection at the point X .

Dykstra's correction term plays a critical role in ensuring convergence to the true projection of A onto $\mathcal{S}_+^n \cap \mathcal{U}^n$, even in cases where classical alternating projections may fail to converge or may oscillate. This convergence is guaranteed by the convexity and non-empty intersection of $\mathcal{S}_+^n$ and $\mathcal{U}^n$, as discussed in Chapter 3.

The convergence rate for Dykstra's algorithm is linear when the sets are subspaces, suggesting at best linear convergence for this problem due to the convex nature of $\mathcal{S}_+^n$ and $\mathcal{U}^n$. Nevertheless, numerical experiments demonstrate the practical effectiveness of Algorithm 1, with convergence achieved within a reasonable number of iterations for problems in financial applications [7].

4.1.3 Exploiting Problem Specifics and Computational Efficiency

To deepen our understanding of the algorithm's behavior, we reference two key results by Higham [7, Thm. 3.4, Cor. 3.5] that connect to the problem setup in the previous chapters.

Theorem 4.1.1 (Structure of Iterates in Algorithm 1). *Suppose $A = A^\top$ has diagonal elements $a_{ii} \geq 1$ and let W be a diagonal matrix. Let $Y_k = \mathcal{P}_{\mathcal{U}^n}(X_k) = X_k + D_k$, where D_k is diagonal (in view of Theorem 3.6.2). Then, in Algorithm 1,*

$$R_k = A + \Delta_k, \tag{4.1}$$

where the diagonal matrix

$$\Delta_k = \sum_{i=1}^{k-1} D_i$$

is negative semidefinite.

Proof. We begin by noting that the algorithm is initialized with $R_1 = A$. Using the recursive formula from Algorithm 1, we derive the expression for R_{k+1} as follows:

$$\begin{aligned} R_{k+1} &= Y_k - \Delta S_k = \mathcal{P}_{\mathcal{U}^n}(X_k) - \Delta S_k \\ &= X_k + D_k - (X_k - R_k) \\ &= R_k + D_k. \end{aligned}$$

By induction, starting from $R_1 = A$, the general form for R_k becomes:

$$R_k = A + \sum_{i=1}^{k-1} D_i = A + \Delta_k,$$

which establishes equation (4.1) from the theorem statement.

To demonstrate that Δ_k is negative semidefinite, we analyze the diagonal elements of the iterates. Since $Y_k = \mathcal{P}_{\mathcal{U}^n}(X_k)$ enforces unit diagonal entries, we have $\text{diag}(Y_k) = e$. Substituting the expression for Y_k , we obtain:

$$\begin{aligned} e &= \text{diag}(Y_k) = \text{diag}(X_k + D_k) \\ &= \text{diag}(\mathcal{P}_{\mathcal{S}^n}(R_k) + D_k) \\ &\geq \text{diag}(R_k + D_k), \\ &= \text{diag}(A + \Delta_{k+1}), \\ &\geq e + \text{diag}(\Delta_{k+1}), \end{aligned}$$

this implies:

$$\text{diag}(\Delta_{k+1}) \leq 0,$$

indicating that Δ_{k+1} is a diagonal matrix with nonpositive entries, and thus $\Delta_{k+1} \preceq 0$. $\square$

Remark 4.1.1. *In particular, if A has t nonpositive eigenvalues, then so does each R_k , as adding a negative semidefinite diagonal matrix cannot increase eigenvalues. As a result, the projection $X_k = \mathcal{P}_{\mathcal{S}_+^n}(R_k)$ will have at least t zero eigenvalues. This observation underpins Corollary 4.1.1 and enables a practical advantage: only the top $n - t$ eigenvalues and eigenvectors need to be computed, reducing cost in eigensolvers. This is especially relevant in high-dimensional financial settings, where empirical correlation matrices often exhibit low effective rank due to redundancy or missing data.*

Corollary 4.1.1 (Computational Efficiency). *Let $A = A^\top$ be a symmetric matrix with diagonal elements $a_{ii} \geq 1$ and t nonpositive eigenvalues, and let W be a diagonal matrix. In Algorithm 1, the intermediate matrix R_k has at least t nonpositive eigenvalues, and the projection $X_k = \mathcal{P}_{\mathcal{S}_+^n}(R_k)$ onto the positive semidefinite cone has at least t zero eigenvalues for all iterations k .*

By letting $k \rightarrow \infty$ in Corollary 4.1.1, we recover the second part of Theorem 3.5.1 from Chapter 3, which states that the nearest correlation matrix has at least t zero eigenvalues. This result is particularly significant for large-scale problems where the matrix dimension n is large, as it enables computational optimizations in Algorithm 1.

When the number of nonpositive eigenvalues t is large (i.e., A is highly rank-deficient), the computation of the projection $\mathcal{P}_{\mathcal{S}_+^n}(R_k)$ can be significantly optimized. From Theorem 3.6.1, the projection is given by:

$$(W^{1/2}R_kW^{1/2})_+ = \sum_{\lambda_i > 0} \lambda_i q_i q_i^\top,$$

where λ_i and q_i are the eigenvalues and corresponding eigenvectors of $W^{1/2}R_kW^{1/2}$. Instead of computing the full eigensystem of $W^{1/2}R_kW^{1/2}$, which is computationally intensive for large n , it suffices to compute only the largest $n - t$ positive eigenvalues and their eigenvectors ($n - t \ll n$). This partial computation can be efficiently performed using methods such as [3, pp. 227 ff.]:

- The Lanczos iteration, which targets the largest eigenvalues of a symmetric matrix.
- Orthogonal reduction of R_k to tridiagonal form, followed by the bisection method and inverse iteration to extract the required eigenvalues and eigenvectors.

For example, in a finance application with a 1399-dimensional stock data matrix, where A had 1245 nonpositive eigenvalues ($t \geq 1245$), the nearest correlation matrix had a rank of at most 154. Using LAPACK's `dsyevr` to compute only the largest eigenvalues reduced computation time from 2 hours 45 minutes to 37 minutes [7].

When t is small relative to n ($t \ll n$), similar computational savings are achievable. In this case, the projection can be computed as:

$$(W^{1/2}R_kW^{1/2})_+ = W^{1/2}R_kW^{1/2} - \sum_{\lambda_i \leq 0} \lambda_i q_i q_i^\top,$$

where the number of nonpositive eigenvalues is estimated and dynamically adjusted during iterations to ensure accuracy. This approach avoids computing the full eigensystem while maintaining numerical stability.

4.2 Accelerated Method for the Nearest Correlation Matrix Problem

Section 4.1 introduced the Modified Alternating Projections as an effective approach for solving the Nearest Correlation Matrix (NCM) problem. While this method guarantees convergence to the unique solution, its linear convergence rate often results in a large number of iterations, especially for large-scale problems. To address this limitation, we explore the application of Anderson acceleration, a technique designed to enhance the convergence of

fixed-point iterations, as proposed by Higham and Strabić [8]. This section outlines the fixed-point interpretation of the modified alternating projections method, presents the Anderson acceleration algorithm, and discusses its convergence properties based on insights from the literature.

4.2.1 Fixed-Point Interpretation

The alternating projections method, as presented in Algorithm 1, can be reformulated as a fixed-point iteration to facilitate the application of Anderson acceleration. Specifically, we define the iteration in terms of a composite state matrix that encapsulates the two primary iterates of the algorithm: Y_k , the matrix projected onto $\mathcal{U}^n$, and ΔS_k , the correction term introduced by Dykstra's method. Let:

$$Z_k = \begin{bmatrix} Y_k \\ \Delta S_k \end{bmatrix} \in \mathbb{R}^{n \times 2n},$$

where both Y_k and ΔS_k are $n \times n$ symmetric matrices. The iteration in Algorithm 1 can then be expressed as a fixed-point problem of the form:

$$Z_{k+1} = g(Z_k),$$

where the function $g: \mathbb{R}^{n \times 2n} \rightarrow \mathbb{R}^{n \times 2n}$ is defined by the steps of Algorithm 1. To make this explicit, we recast Algorithm 1 into a fixed-point form, as described in [8, Alg. 4].

4.2.2 Fixed-Point Form of Algorithm 1

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, this algorithm computes the nearest correlation matrix Y to A . It requires a convergence tolerance tol .

Algorithm 2 Fixed-Point Form of Algorithm 1

- 1: **Input:** Symmetric matrix $A \in \mathbb{R}^{n \times n}$, tolerance $\text{tol} > 0$
 - 2: **Output:** Approximate correlation matrix $Y_k \in \mathbb{R}^{n \times n}$
 - 3: **Initialize:** $\Delta S_0 \leftarrow 0, Y_0 \leftarrow A$
 - 4: **for** $k = 1, 2, \dots$ **do**
 - 5: Compute $[Y_k, \Delta S_k] \leftarrow g(Y_{k-1}, \Delta S_{k-1})$, as follows:
 - 6: $R_k \leftarrow Y_{k-1} - \Delta S_{k-1}$
 - 7: $X_k \leftarrow \mathcal{P}_{S_+^n}(R_k)$
 - 8: $\Delta S_k \leftarrow X_k - R_k$
 - 9: $Y_k \leftarrow \mathcal{P}_{\mathcal{U}^n}(X_k)$
 - 10: **if** $\|Y_k - X_k\|_F \leq \text{tol} \cdot \|Y_k\|_F$ **then**
 - 11: **Return** Y_k
-

In this formulation, the function g maps the pair $(Y_{k-1}, \Delta S_{k-1})$ to $(Y_k, \Delta S_k)$, which corresponds to the update $Z_{k+1} = g(Z_k)$. This fixed-point representation allows the use of acceleration techniques for fixed-point iterations, such as Anderson acceleration.

4.2.3 Anderson Acceleration for Fixed-Point

Anderson acceleration is designed to enhance the convergence of fixed-point iterations. Unlike standard fixed-point iterations of the form $x_{k+1} = g(x_k)$, which rely solely on the current iterate, Anderson acceleration utilizes information from up to m_k previous iterations to compute the next iterate as a linear combination of these quantities. By reformulating the fixed-point problem $x = g(x)$ as finding a zero of the function $f(x) = g(x) - x$, Algorithm 3 iteratively seeks a solution to $f(x) = 0$. This approach is particularly relevant for problems such as the Nearest Correlation Matrix (NCM), where accelerated convergence can significantly improve computational efficiency [8, Alg. 2].

Algorithm 3 Anderson Acceleration

- 1: **Input:** Initial vector $x_0 \in \mathbb{R}^n$, history parameter $m \geq 1$, function $f(x) = g(x) - x$
 - 2: **Output:** Sequence $\{x_k\}$ converging to a fixed point of f
 - 3: **Initialize:** $x_1 \leftarrow x_0 + f(x_0)$
 - 4: **for** $k = 1, 2, \dots$ until convergence **do**
 - 5: $m_k \leftarrow \min(m, k)$
 - 6: $f_k \leftarrow f(x_k)$
 - 7: Define:

$$\Delta x_i \leftarrow x_{i+1} - x_i, \quad f_i \leftarrow f(x_i)$$

$$\Delta f_i \leftarrow f_{i+1} - f_i$$

$$X_k \leftarrow [\Delta x_{k-m_k}, \dots, \Delta x_{k-1}]$$

$$F_k \leftarrow [\Delta f_{k-m_k}, \dots, \Delta f_{k-1}]$$
 - 8: $\gamma^{(k)} \leftarrow \arg \min_{\gamma} \|f_k - F_k \gamma\|_2$
 - 9: $x_k \leftarrow x_k - X_k \gamma^{(k)}$
 - 10: $f_k \leftarrow f_k - F_k \gamma^{(k)}$
 - 11: $x_{k+1} \leftarrow x_k + f_k$
-

The computational overhead of Anderson acceleration, relative to unaccelerated fixed-point iterations, includes an additional cost of at most $\frac{3m^2}{2} + 12mn + 3n$ floating-point operations per iteration. Storage requirements increase by $2n^2m$ elements to accommodate the history of iterates and function values. For the NCM problem, where iterates are represented as vectors of size $2n^2$ and m is typically small (e.g., $m \leq 5$), the additional cost is $O(n^2)$, which is negligible for large n . The dominant computational expense often arises from the eigenvalue decomposition required in the underlying alternating projections method.

While Anderson acceleration lacks general convergence guarantees for nonlinear problems, it has demonstrated empirical success in applications such as electronic structure computations and domain decomposition methods. Under specific conditions, local linear convergence has been established for nonlinear problems. Numerical experiments for the NCM problem indicate that Anderson acceleration produces results closely aligned with those of the unaccelerated alternating projections method, underscoring its practical utility despite the absence of universal theoretical guarantees.

4.2.4 Anderson Acceleration for the Nearest Correlation Matrix

Algorithm 4 enhances the modified alternating projections method by applying Anderson acceleration (Algorithm 3) to compute the nearest correlation matrix to a given symmetric matrix $A \in \mathbb{R}^{n \times n}$. This approach addresses the challenge of identifying a symmetric positive semidefinite matrix with unit diagonal that minimizes the distance to A , typically in the Frobenius norm. Such problems are prevalent in financial applications and data analysis, where empirical matrices may be indefinite due to missing or asynchronous observations.

Algorithm 4 leverages Anderson acceleration to accelerate the fixed-point formulation of the alternating projections method [8, Alg. 3.2].

Algorithm 4 Anderson Accelerated Nearest Correlation Matrix

- 1: **Input:** Symmetric matrix $A \in \mathbb{R}^{n \times n}$, tolerance $\text{tol} > 0$
 - 2: **Output:** Approximate nearest correlation matrix $Y \in \mathbb{R}^{n \times n}$
 - 3: Run Algorithm 3 on the function $f: \mathbb{R}^{2n^2} \rightarrow \mathbb{R}^{2n^2}$ given by

$$f(z) = \text{vec}(\widetilde{g}(Z) - Z),$$
 where $z_k = \text{vec}(Z_k)$, $Z_k = (Y_k, \Delta S_k) \in \mathbb{R}^{n \times 2n}$
 and $[X_k, \widetilde{g}(Z_k)] = g(Z_k)$, for the function g defined by Algorithm 2
 - 4: Terminate the iteration when

$$\frac{\|Y_k - X_k\|_2}{\|Y_k\|_2} \leq \text{tol}$$
 - 5: Denote the result by z^*
 - 6: $Y \leftarrow \text{unvec}(z^*)$
-

Algorithm 4 builds on the modified alternating projections method (Algorithm 2), which iteratively projects onto $\mathcal{S}_+^n$ and $\mathcal{U}^n$. While Algorithm 2 is straightforward to implement and adaptable to additional constraints, its linear convergence rate often requires numerous iterations. Algorithm 4 mitigates this by applying Anderson acceleration, aiming to reduce iteration counts and computation time.

The computational cost of Algorithm 2 is dominated by the projection $\mathcal{P}_{\mathcal{S}_+^n}(R_k)$, requiring an eigenvalue decomposition at approximately $10n^3$ flops (or $\frac{14n^3}{3}$ with tridiagonalization and bisection/inverse iteration). Algorithm 4 incurs an additional cost from Anderson acceleration of at most $\frac{3m^2}{2} + 24mn^2 + 6n^2$ flops per iteration. For small m (e.g., $m \leq 5$), this cost is $O(n^2)$, negligible compared to the $O(n^3)$ cost of eigenvalue decomposition for large n . Storage increases by $2n^2m$ elements to store the history of iterates.

While Algorithm 2 guarantees convergence to the unique nearest correlation matrix (assuming no fixed off-diagonal elements), Algorithm 4 lacks general convergence guarantees

for nonlinear problems. However, numerical experiments demonstrate that Algorithm 4 significantly reduces iteration counts and computation time compared to Algorithm 2, with computed matrices agreeing closely with unaccelerated results within a small multiple of the tolerance. The algorithm is particularly effective for challenging problems, such as those with fixed elements or constraints on the smallest eigenvalue.

The integration of Anderson acceleration in Algorithm 4 enhances the efficiency of the alternating projections method, making it well-suited for financial and data analysis applications. The ability to leverage historical iterates reduces the computational burden, particularly for large-scale problems. However, the absence of theoretical convergence guarantees for nonlinear problems necessitates cautious application. Further research into conditions ensuring convergence or the development of hybrid methods could improve the robustness and applicability of Algorithm 4, especially for problems with complex constraints.

CHAPTER 5

MISSING DATA MECHANISMS IN STATISTICAL ANALYSIS

The study of missing data mechanisms forms a critical foundation for statistical methodology, providing a structured approach to understanding how incomplete datasets influence research outcomes. This chapter discusses the three main mechanisms; Missing Completely at Random (MCAR), Missing at Random (MAR), and Not Missing at Random (NMAR). A clear grasp of these mechanisms is essential for selecting appropriate techniques to handle missing data, ensuring robust and unbiased statistical inferences [20].

5.1 Missing Data Mechanisms

This section details the three main missing data mechanisms MCAR, MAR, and NMAR providing clear definitions and examples in financial applications.

5.1.1 Missing Completely at Random (MCAR)

Definition: Missing Completely at Random occurs when the likelihood of missing data is entirely independent of both observed and missing data values. The missingness is a purely random process, unaffected by any variable in the dataset, ensuring no systematic bias if handled correctly. The probability of missingness is expressed as:

$$p(R \mid \phi),$$

where R is the missing data indicator ($R = 1$ for observed data, $R = 0$ for missing data), and ϕ governs the missingness process (describes the relationship between R and data). Under MCAR, this probability does not depend on the observed data (Y_{obs}) or missing data (Y_{mis}).

Practical Example: In a financial time-series dataset, such as foreign exchange rates, missing data may occur due to random technical errors, like a system failure that omits recordings on certain days, unrelated to the actual exchange rates or other measured variables. Other real-world instances include random entry loss in Bloomberg feeds, exchange outages (e.g., NYSE, July 2015), or deliberate uniform subsampling applied for weekly correlation analysis. These scenarios illustrate how missingness can arise either unintentionally through data feed disruptions or intentionally through data reduction techniques, both impacting the completeness of the observed dataset.

5.1.2 Missing at Random (MAR)

Definition: Missing at Random indicates that the probability of missing data depends on observed data (Y_{obs}) but is independent of the missing data (Y_{mis}) after accounting for observed variables. Despite its name, MAR does not imply randomness in a colloquial sense; rather, missingness is systematically related to measured variables, allowing conditional independence. The probability of missingness is:

$$p(R \mid Y_{\text{obs}}, \phi),$$

indicating that R depends on Y_{obs} but not Y_{mis} , given ϕ .

Practical Example: In a study of stock returns, missing data might arise because certain firms fail to report earnings due to regulatory exemptions tied to firm size, a measured

variable. If firm size explains the missingness, and the missing returns are unrelated to their actual values after controlling for size, the data are MAR.

5.1.3 Not Missing at Random (NMAR)

Definition: Not Missing at Random occurs when the probability of missing data depends on the missing values themselves (Y_{mis}), even after accounting for observed data (Y_{obs}). This mechanism reflects a direct relationship between missingness and unobserved data, posing significant analytical challenges. The probability of missingness is:

$$p(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \phi),$$

where R depends on both Y_{obs} and Y_{mis} , governed by ϕ .

Practical Example: In financial time-series data, missing interest rate indices might occur because extreme values are not reported due to market sensitivity, where the missingness is directly tied to the unreported values themselves. Other real-world cases include the exclusion of low-magnitude correlations (e.g., $|\rho| < 0.2$) from published datasets, regulatory omissions such as bank–insurance correlations withheld under the Volcker Rule, or data loss caused by asset delisting events that remove certain securities from ongoing reporting. These scenarios illustrate how missingness can be systematically related to the underlying data values or regulatory decisions, rather than occurring at random.

5.2 Missing Data Simulation

To evaluate the performance of correlation matrix estimation methods under incomplete data scenarios, we implement simulation procedures for two fundamental missing data mechanisms: Missing Completely at Random (MCAR) and Not Missing at Random (NMAR). These simulations, designed with deletion probabilities ranging from 5% to 50% ($p \in [0.05, 0.50]$), enable us to assess the robustness and accuracy of the algorithms in financial data analysis, where missing data is prevalent. The MCAR mechanism produces a scattered pattern of missing entries, while NMAR results in a structured pattern, reflecting dependence on unobserved data values.

5.2.1 Missing Completely at Random (MCAR) Simulation

The MCAR mechanism assumes that the probability of missingness is independent of both observed and unobserved data values. Missing entries are introduced into a symmetric matrix by randomly selecting off-diagonal elements with a deletion probability $p \in [0.05, 0.50]$, ensuring symmetry by setting corresponding pairs (i, j) and (j, i) to NaN. This process yields a scattered pattern of missing entries, as the likelihood of missingness is uniform across all off-diagonal positions, independent of the values of the entries of the matrix. Visually, the missing entries appear randomly dispersed, resembling noise without discernible structure.

To illustrate, consider a 6×6 symmetric matrix with 12 missing off-diagonal entries (corresponding to 6 symmetric pairs) generated under the MCAR mechanism with $p = 0.4$, with missing entries denoted by NaN:

$$A = \begin{bmatrix} 1 & \text{NaN} & 0.12 & 0.08 & \text{NaN} & 0.31 \\ \text{NaN} & 1 & \text{NaN} & 0.15 & 0.26 & \text{NaN} \\ 0.12 & \text{NaN} & 1 & 0.17 & \text{NaN} & 0.22 \\ 0.08 & 0.15 & 0.17 & 1 & 0.29 & \text{NaN} \\ \text{NaN} & 0.26 & \text{NaN} & 0.29 & 1 & 0.19 \\ 0.31 & \text{NaN} & 0.22 & \text{NaN} & 0.19 & 1 \end{bmatrix}.$$

This matrix maintains symmetry (if $A_{ij} = \text{NaN}$, then $A_{ji} = \text{NaN}$) and unit diagonal entries, characteristic of correlation matrices. The 12 missing values are randomly distributed, reflecting the scattered pattern typical of MCAR.

5.2.2 Not Missing at Random (NMAR) Simulation

The NMAR mechanism assumes that the probability of missingness depends on the unobserved data values, rendering the missing data process non-ignorable. Missing entries are introduced by targeting off-diagonal elements with larger absolute values, using a threshold parameter adjusted to achieve a deletion proportion within $[0.05, 0.50]$, and setting corresponding pairs (i, j) and (j, i) to NaN. This results in a structured pattern of missing entries, as missingness is systematically tied to the magnitude of the correlations. Visually, the missing entries tend to cluster around positions corresponding to stronger relationships, reflecting the underlying data structure rather than random dispersion.

To illustrate, consider a 6×6 symmetric matrix with 12 missing off-diagonal entries (6 symmetric pairs) under NMAR, with a threshold $\tau = 0.2$ calibrated to achieve approximately 40% missingness:

$$A = \begin{bmatrix} 1 & \text{NaN} & 0.12 & 0.08 & \text{NaN} & \text{NaN} \\ \text{NaN} & 1 & \text{NaN} & 0.15 & \text{NaN} & \text{NaN} \\ 0.12 & \text{NaN} & 1 & 0.17 & \text{NaN} & 0.22 \\ 0.08 & 0.15 & 0.17 & 1 & \text{NaN} & \text{NaN} \\ \text{NaN} & \text{NaN} & \text{NaN} & \text{NaN} & 1 & 0.19 \\ \text{NaN} & \text{NaN} & 0.22 & \text{NaN} & 0.19 & 1 \end{bmatrix}.$$

This matrix preserves symmetry and unit diagonal entries, with 12 missing values corresponding to 6 symmetric pairs where $|A_{ij}| > 0.2$. The structured pattern is evident, as missing entries are concentrated in positions with larger correlations, reflecting the NMAR dependence on unobserved data values.

Remark 5.2.1. *The MCAR and NMAR simulations, exemplified by the matrices above, provide distinct environments to test the algorithms under ignorable and non-ignorable missing data mechanisms, respectively, as discussed in Section 5.2. The scattered pattern of MCAR aligns with financial scenarios where missingness results from random external factors, while the structured pattern of NMAR addresses cases tied to the latent structure of the data, such as unobserved market conditions. By employing deletion probabilities of 5% to 50%, we capture a range of missingness scenarios, enhancing the robustness of our analysis.*

CHAPTER 6

PROJECTION METHODS AND MISSING-DATA ANALYSIS

6.1 Experimental Setup

We evaluated the methods using carefully constructed synthetic matrices and real-world financial data.

Synthetic Correlation Matrices.

- **Higham’s 4×4 Test Case [7]:**

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

It lacks unit diagonal entries but satisfies the positive semidefiniteness condition, making it a suitable benchmark for evaluating algorithms that enforce both key properties of correlation matrices: unit diagonal and positive semidefiniteness.

- **Toeplitz Matrix (6×6):**

$$A_{ij} = 0.9^{|i-j|}, \quad i, j = 1, \dots, 6$$

$$A = \begin{pmatrix} 1 & 0.9 & 0.81 & 0.729 & 0.6561 & 0.5905 \\ 0.9 & 1 & 0.9 & 0.81 & 0.729 & 0.6561 \\ 0.81 & 0.9 & 1 & 0.9 & 0.81 & 0.729 \\ 0.729 & 0.81 & 0.9 & 1 & 0.9 & 0.81 \\ 0.6561 & 0.729 & 0.81 & 0.9 & 1 & 0.9 \\ 0.5905 & 0.6561 & 0.729 & 0.81 & 0.9 & 1 \end{pmatrix}$$

This matrix models decaying correlations typical in financial time-series.

Real-World Financial Data. We analyze a comprehensive equity dataset with the following characteristics, which can be viewed in the corresponding GitHub repository [22]:

- **Stock Selection:**

- A dataset comprising 550 stocks from global equity markets spanning 2020–2025, containing approximately 768,900 observations.
- Balanced sector representation: 50 stocks randomly selected from each of 11 sectors.
- Equal weighting scheme applied (no market-cap bias).

- **Return Calculation:** Let P_T denote the asset price at time T . Then the log return is computed as:

$$r_T = \ln \left(\frac{P_T}{P_{T-1}} \right).$$

- **Correlation Estimation:** Full-period Pearson correlation coefficients are computed as:

$$\rho_{ij} = \frac{\langle \mathbf{r}_i - \bar{r}_i, \mathbf{r}_j - \bar{r}_j \rangle}{\|\mathbf{r}_i - \bar{r}_i\|_2 \|\mathbf{r}_j - \bar{r}_j\|_2}$$

where:

- $\mathbf{r}_i$ denotes the full return series for asset i ,
- $\bar{r}_i$ is the mean return over the full time period.

Table 6.1: Counts of missing data entries under MCAR and NMAR mechanisms across varying deletion rates in a 550×550 stock correlation matrix (2020–2025).

Missing Rate(%)	MCAR	NMAR
5	38,192	38,418
10	76,538	76,835
20	153,390	153,670
30	230,985	230,505
40	307,458	307,340
50	383,988	384,175

6.1.1 Evaluation Metrics

Performance is assessed using three key metrics:

- **Iterations:** Total projection steps until convergence.
- **Computation Time:** Wall-clock time in seconds.
- **Reconstruction Error:** $\|A - X\|_F$.

6.1.2 Implementation Details

- **Software Environment:**

- Implemented in Julia 1.11.6.
- Core operations using `LinearAlgebra.jl`.
- Random number generation with fixed seed (RNG seed = 1, 5, 10, 20, 30, 40, 50).

- **Hardware Specifications:**

- Processor: AMD Ryzen 5 5625U with Radeon Graphics (2.30 GHz).
- Memory: 8.00 GB DDR4 (7.35 GB usable).
- Architecture: 64-bit x64-based system.

6.2 Comparative Analysis of Projection Methods

Building on the methodology introduced in Chapter 4, we compare the modified alternating projections (MAP) and Anderson acceleration (AA) methods for estimating correlation matrices on both synthetic matrices and real-world financial datasets in Section 6.1.

6.2.1 Performance Results

Table 6.2 summarizes the performance metrics for MAP and AA across six test cases: Higham’s 4×4 matrix (Missing = 0%), the 6×6 Toeplitz matrix (Missing = 10%, corresponding to 4 missing entries), and the real-world 550×550 correlation matrix (Missing = 1%, corresponding to 7,000+ and 25%, corresponding to 192,000+ missing entries).

Table 6.2: Performance comparison between Modified Alternating Projections (MAP) and Anderson Acceleration (AA) across six test cases.

Example	Method	Iterations	Time (s)	$\ A - X\ _F$
Higham 4×4	MAP	19	0.00104	2.133
	AA	3	0.00601	2.251
Toeplitz 6×6	MAP	27	0.00159	0.369
	AA	4	0.00477	0.402
Real-world 550×550 (1% MCAR)	MAP	13	0.656	0.116
	AA	3	0.686	0.124
Real-world 550×550 (1% NMAR)	MAP	19	0.914	0.0392
	AA	3	1.18	0.0412
Real-world 550×550 (25% MCAR)	MAP	84	6.18	4.832
	AA	10	2.26	5.072
Real-world 550×550 (25% NMAR)	MAP	72	4.75	3.558
	AA	7	1.88	3.734

Remark 6.2.1. *Anderson Acceleration (AA) was selected for missing-data reconstruction of correlation matrices due to its significant reduction in iteration counts compared to Modified Alternating Projections (MAP). AA demonstrates markedly superior convergence, particularly at higher missing-data proportions. AA requires 88% fewer iterations in the 25% MCAR*

scenario (10 vs 84 iterations) and 90% fewer iterations in the 25% NMAR scenario (7 vs 72 iterations).

Both AA and MAP produce minimum eigenvalues that are effectively zero within numerical precision, with any deviations arising from floating-point arithmetic; AA tends to yield slightly positive eigenvalues, whereas MAP tends to yield slightly negative eigenvalues which can be fixed by imposing a lower bound on the smallest eigenvalue because singularity is an issue in applications where the inverse of the correction matrix is needed [8]. For the 4×4 Higham's matrix, AA yields a minimum eigenvalue of 0.1275 compared to MAP's -5.1×10^{-9} , and for the 6×6 Toeplitz matrix AA yields 0.00022 compared to MAP's -5.07×10^{-9} . For large matrices, the real-world 550×550 matrix under 1% MCAR AA yields 2.78×10^{-4} compared to MAP's -1.70×10^{-12} ; under 1% NMAR AA yields 1.67×10^{-4} compared to MAP's -2.87×10^{-12} ; under 25% MCAR AA yields 1.43×10^{-7} compared to MAP's -7.73×10^{-12} ; and under 25% NMAR AA yields 2.49×10^{-6} compared to MAP's -6.17×10^{-12} . AA reliability eliminates the need for post-processing corrections that MAP occasionally requires, a critical advantage for downstream financial applications such as portfolio optimization and Monte Carlo simulations.

Beyond iteration counts and eigenvalue behavior, AA shows increasingly favorable wall-clock performance as missingness rises. While AA is initially slower than MAP for small matrices ($5.8\times$ slower for the 4×4 Higham matrix and $3\times$ slower for the 6×6 Toeplitz matrix), the runtime gap narrows and reverses for larger matrices at higher missingness levels. For the real-world 550×550 matrix, AA is only 4% slower at 1% MCAR (0.686 s vs 0.656 s) and AA is only 23% slower at 1% NMAR (1.18 s vs 0.914 s), but becomes 63% faster at 25% MCAR (2.26 s vs 6.18 s) and 60% faster at 25% NMAR (1.88 s vs 4.75 s). These results indicate that AA scales more effectively with increasing matrix size and missingness, both in convergence and runtime.

MAP's speed advantage at low missingness stems from its highly optimized Julia implementation, leveraging high-performance linear algebra libraries such as Eigen and LAPACK. Accuracy between the two methods remains comparable, with MAP achieving marginally lower Frobenius norm errors across all cases: 5% for the Higham matrix (2.133 vs 2.251), 8% for the Toeplitz matrix (0.369 vs 0.402), 7% for the real-world matrix under 1% MCAR (0.116 vs 0.124), 5% under 1% NMAR (0.0392 vs 0.0412), and 5% at 25% MCAR (4.832 vs 5.072) and 25% NMAR (3.558 vs 3.734). Both methods maintain robust performance up to 50% missingness under MCAR and NMAR mechanisms. A key advantage of projection-based methods over all other approaches for computing the nearest correlation matrix is their flexibility allowing seamless incorporation of fixed-element and eigenvalue constraints [8].

6.3 Analysis of Missing Mechanisms

We apply the Anderson acceleration (AA) to assess its performance under both MCAR and NMAR mechanisms.

Remark 6.3.1. *Anderson acceleration offers markedly faster convergence and often reduced computational time, particularly under Not Missing At Random (NMAR) data mechanisms. Crucially, it consistently yields well-conditioned correlation matrices, maintaining positive semidefiniteness with minimum zero eigenvalues across a wide range of missing-data levels and mechanisms. This structural integrity is vital for subsequent statistical analyses, making it a highly reliable and robust choice for estimating correlation matrices in practical applications.*

Table 6.3: AA performance under MCAR and NMAR mechanisms.

Mechanism	Missing (%)	Iterations	Time(s)	$\ A - X\ _F$
MCAR	5 %	4	1.07	0.827
MCAR	10 %	7	1.41	1.647
MCAR	20 %	7	1.50	3.369
MCAR	30 %	30	5.46	6.342
MCAR	40 %	46	7.87	9.998
MCAR	50 %	55	9.06	14.845
NMAR	5 %	5	1.18	0.921
NMAR	10 %	5	1.28	1.703
NMAR	20 %	7	1.40	2.717
NMAR	30 %	10	1.83	4.961
NMAR	40 %	38	6.51	8.211
NMAR	50 %	72	12.8	12.881

CHAPTER 7

CONCLUSIONS

7.1 Summary of Findings

This thesis has systematically investigated the nearest correlation matrix (NCM) problem in the context of financial applications, with a particular focus on addressing the challenges posed by missing data. The primary objective was to state and compare efficient algorithms for computing the nearest correlation matrix while handling missing data mechanisms, specifically Missing Completely at Random (MCAR) and Not Missing at Random (NMAR). Through a comprehensive analysis, we evaluated two key methods: the Modified Alternating Projections (MAP) method and the Anderson Acceleration (AA) method, with an emphasis on their computational efficiency, convergence speed, accuracy, robustness, and scalability.

Chapter 3 established the theoretical foundation by defining the NCM problem as a convex optimization problem, minimizing the weighted Frobenius norm distance between an empirical matrix and a valid correlation matrix satisfying symmetry, positive semidefiniteness, and unit diagonal constraints. The mathematical characterizations, including normal cones and projection operators, provided the basis for the algorithmic developments in Chapter 4. The MAP method, enhanced by Dykstra's correction, ensured stable convergence by iteratively projecting onto the convex sets $\mathcal{S}_+^n$ (symmetric positive semidefinite matrices) and $\mathcal{U}^n$ (symmetric matrices with unit diagonal). The AA method, by reformulating the

problem as a fixed-point iteration and leveraging historical iterates, significantly improved the convergence speed.

Chapter 5 explored missing data mechanisms, detailing the definitions and implications of MCAR, MAR, and NMAR. By simulating the MCAR and NMAR scenarios, we evaluated their impact on the estimation of the correlation matrix. The simulations revealed that MCAR produces scattered missing patterns, while NMAR results in structured missing patterns.

The experimental analysis in Chapter 6 demonstrated that the AA method, configured with $m = 4$, performed consistently or better than the MAP implementation, which uses optimized Eigen routines, as the matrix size and missing data fraction increased. Across various test cases including Higham’s 4×4 , the 6×6 Toeplitz, and a real-world 550×550 correlation matrix, AA required fewer outer iterations and consistently converged to spectra whose minimum eigenvalues reached zero within numerical precision. As missingness increased, AA maintained robust positive semidefinite reconstructions. Under the NMAR structured missingness mechanism, AA achieved reconstructions marginally closer to the true matrix in Frobenius norm compared to MAP.

7.2 Implications for Financial Modeling

The experimental findings carry significant implications for financial modeling, particularly in portfolio optimization and risk management. Accurate correlation matrices are essential for capturing asset interdependencies, yet financial time series frequently contain gaps due to market holidays, reporting delays, and regulatory constraints. These missing data patterns can substantially degrade matrix estimation quality, introducing instability that propagates through downstream analyses.

Our results demonstrate that AA effectively addresses these challenges. Across both MCAR and NMAR missing data mechanisms, AA consistently produced positive semidefinite matrices with stable eigenvalue spectra, even at missingness levels reaching 50%. In contrast, MAP allowed eigenvalues to drift into negative ranges under comparable conditions, compromising matrix validity. This structural reliability translates directly into more dependable inputs for financial analysis, reducing the risk of suboptimal investment decisions stemming from ill-conditioned or near-singular correlation matrices.

Beyond technical accuracy, AA offers compelling practical advantages. Its faster convergence and lower computational requirements make it well-suited for large-scale financial datasets where both efficiency and precision are paramount. Together, these characteristics position AA as a robust foundation for enhanced risk management practices and more reliable portfolio optimization outcomes.

7.3 Limitations

Our experimental evaluation revealed performance degradation at higher missingness levels that affected both AA and MAP across missingness mechanisms. From 30% missingness, computational demands increased substantially for both methods. More critically, matrix quality deteriorated: reconstructed matrices exhibited increased singularity, and higher Frobenius norms indicated significant deviations from the original correlation structure. Across missingness levels, NMAR scenarios generally preserved better matrix quality than MCAR, as evidenced by fewer zero eigenvalues and more stable spectral properties. However, this relative advantage diminished at 40% and 50% missingness, where both mechanisms produced increasingly degraded reconstructions. These findings suggest that beyond a certain missingness threshold approximately 30-40% in our experiments imputation reliabil-

ity inherently declines regardless of the underlying missingness mechanism or reconstruction method employed.

The exclusion of MAR mechanism, due to the complexity of modeling dependencies between observed and missing data, restricts the generalization of our findings to real-world financial data, where such dependencies are common.

Additionally, while the AA method demonstrated empirical success, its lack of universal convergence guarantees for nonlinear problems requires cautious application, especially in cases with complex constraints or extreme missingness patterns beyond those evaluated.

Finally, the reliance on synthetic matrices and a single real-world dataset may not fully represent the diversity of financial data structures encountered in practice, potentially limiting the broader applicability of the reported performance metrics.

7.4 Future Research Directions

Building on the findings of this study, future research should directly address the current limitations and explore opportunities to strengthen the practical relevance of the proposed methodology. A key direction involves rigorously incorporating MAR mechanisms, which are common in real-world financial datasets. This would require explicitly modeling the dependencies between observed and unobserved data, thereby extending the current framework beyond the MCAR and NMAR mechanisms analyzed in this thesis.

Given the observed behavior of AA at higher levels of missingness (30–50%), where increasing singularity and larger Frobenius norm deviations were noted, a more detailed investigation into the effect of missing data on the eigenvalue distributions of correlation matrices is warranted. Understanding these effects in the context of portfolio optimization, where matrix conditioning and stability are critical, would provide valuable practical insights.

Beyond mean-variance optimization, extending the analysis to other areas of finance, such as fixed-income securities and derivative portfolios, would further test the generalizability of the results and reveal how missing data affects the shape of the efficient frontier across different asset classes.

Finally, to fully assess the scalability and practical limits of the AA method, future studies should evaluate its performance on substantially larger financial datasets, including settings with millions of observations and thousands of assets significantly exceeding the 768,900 observations and 550 assets used in this thesis. Such large-scale experiments would help establish the computational efficiency and robustness of the method in high-dimensional, real-world financial environments.

7.5 Concluding Remarks

This thesis advances the understanding of nearest correlation matrix computation in the presence of missing data, offering practical insights for financial practitioners. The findings demonstrate that the Anderson Acceleration method emerges as a superior choice, combining computational speed with robustness across varying levels of missingness. This makes it a valuable tool for real-world financial modeling where data quality and computational constraints are pervasive challenges. By systematically addressing the interplay between missing data mechanisms and computational efficiency, this research contributes to the development of reliable and scalable solutions for correlation matrix estimation, with broader implications extending beyond finance to statistical applications requiring high-dimensional covariance analysis.

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